Pre-analysis plan for ManyDaughters Project

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Hypothesis

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

That is, denoting X as a measure of support for gender equality,

(X for respondents with daughters) > (X for respondents without daughters).

H2. Having daughters leads to more left-leaning political preferences.

That is, denoting Y as a measure of left-leaning political preferences,

(Y for respondents with daughters) > (Y for respondents without daughters).

H3. Having daughters leads to more gender equality in the household, as measured by behaviors or outcomes.

That is, denoting Z as a measure of gender equality in the household,

(Z for respondents with daughters) > (Z for respondents without daughters).

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

That is, denoting X as a measure of support for gender equality,

(X for male respondents with daughters − X for male respondents without daughters) >

(X for female respondents with daughters − X for female respondents without daughters)

Variables

(1) *Main predictor of interest: Sex of first child*

kidsex01 coded into (0) son and (1) daughter

(2) Outcome hypothesis 1: Gender equality measured by attitudes

- We use the variables plh0300_v1 to plh0309.
- We exclude three variables, namely "Marriage is a Lifelong Union" (plh0305), "Mariage when child is born" (plh0306), and "Marriage when living with partner" (plh0300*), as they, in our opinion, related too little to gender equality and more related to marriage in general.
- We only take the year 2012, ignoring 2018, because the items in 2018 were only asked when 2012 was missing, but the time break (2012 and 2018) is too long (meaning a child could have been born in between), and only two items are available in 2018.
- We code all items in the same direction, with higher values indicating more gender equality.

- We see a scale with a Cronbach's alpha above 0.70 as sufficiently correlated. In the case of a lower Cronbach's alpha, we delete the least correlated item until the threshold is reached. If there is, however, a clear item that is theoretically and empirically not correlated, we are excluding it based on theory as well. Cronbach's is based on the full sample and not our restricted one, but the missings on the items are excluded when calculating the Cronbach's alpha.
- General handling of missing: We take the individual average of all items if there is an answer on X items (X = total amount/2 +1), meaning that missings on multiple items are possible.
- To make the scales comparable and interpretation more straightforward, the items and scale are transformed to range from 0 to 1 by (scale- r(min)) / (r(max) - r(min)).

(3) *Outcome hypothesis 2: left-leaning political preferences*

- We use the existing variable plh0004.
- We turn the variable around to have (10) as very leftish and (0) being more right.
- The linear scale is kept to see little differences and to use all the available information.
- As the variable is only included in the panel years 2005, 2008, 2014, and 2019, only those years are used.
- We do not use party preference, as only those who say they have a preference are able to fill in the question and we argue that one can have a more left-leaning political preference even if they do not have a party preference.

(4) *Outcome hypothesis 3: gender equality in the household, as measured by behaviors or outcomes*

- We use household tasks other than paid work during the week similar to (Kühhirt, 2012) namely the following variables:
  Childcare during week: pli0044_v3
  Housework during week: pli0043_h
  Errands during week: pli0040
- As couples are compared, we take the latest year in which we have information on both couples' tasks. All tasks have to be filled in as we do not know if the partner potentially is making up for missing tasks by doing other tasks. In the case that one of the partners have a missing, the complete couple is not taken into account.
- Also, they have to have a partner living in the same household.
- To compare equality between partners in the household, the sum of hours on all tasks of one partner is compared to the sum of hours on all tasks of the other partner. We do

so by calculating the share of hours each partner contributes to the total hours in the household. Like this, it is possible to take into account that some households will have fewer hours in total (e.g., when help is provided from outside the household) and we are interested in the share between the couples.

- Last, we transform the variable to have the value 1 as completely equal distribution in the household, meaning each partner does 50% of the tasks, and a value of 0 if one partner does all the unpaid work. In that way of transforming the variables, a contribution of 80:20 is the same as 20:80. More precisely, we calculate it like the following:
    - 1- (|(respondents total hours/(total hours of both couples calculated by summing respondents total hours + partners total hours))-0.5 | *2)
    - In stata code this looks like:

```
gen share = (respondents total hours / (total hours of both couples)) - 0.5
replace share = share*(-1) if share <0
gen equality = 1- (share*2)
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(5) *Interaction variable for hypothesis 4: Sex*

- Sex of respondent using variable "sex"

No control variables as the sex of a child should be relatively random in a country such as Germany, where the sex is only told after abortion is no longer possible.

Inclusion/exclusion criteria

(1) *Include only respondents who have children*. Reason: Those with children and without children are very different in the outcomes, so a comparison would mean controlling for many things (as many characteristics of a person relate to deciding to have children but also to the outcome variables), which are likely not all available in the data. Furthermore, having a daughter rather than a son is relatively random and helps establish a causal relationship more effectively.

(2) *Include only respondents who have one child and exclude them before they have their second child (Select if kidgeb02 > syear // second child born after year of survey)*. We make an exception for twins: if both children are girls then we count them as 'having (one) daughter', if both children are boys then we count them as 'having (one) son'.
Reason for only including respondents whilst they have their first child: With more children, the likelihood of having a daughter is higher. Yet at the same time, the relationship could be

the other way around, that the decision to have more children is influenced by the sex composition of the earlier child(ren) (see, for instance, Mills & Begall, 2010). Put together, the number of children could be both a control variable and a mediator, therefore, to make the causal link stronger, the selection of looking at the first child is made.

The age of the child does not matter to us as we argue that the influence is immediate and does not fade away over time. Also, the location where the child is living is, in our opinion, not relevant.

(3) Exclude respondents for whom we do not have information on the child's sex or on the outcome variables.

(4) *Include the last year of the panel where we have full information, and include each respondent only once*. Reason: To have as few missing values as possible and not to have the respondent multiple times in the regression.

This means that hypothesis H1 and H4 will be on the year 2012 only; Hypothesis 2 will be on the years 2005, 2009, 2014, and 2019; Hypothesis 3 is not specified further, so it could be between the years 2000 and 2020

Exact analysis

(1) Panel analysis is not needed because having a daughter and not a son is relatively random. Further, the hypotheses are not formulated in a panel way. For fixed-effects, one would formulate the hypothesis in terms of how *becoming* a parent to a daughter changes the outcome. This means the formulation indicates a cross-sectional analysis.

(2) Cluster on household with variable hid as it is a household panel, so there is overlap in children and household divisions which we want to account for.

(3) OLS using the reg command.

More precisely, the code for the analysis will be:

H1: reg attitude_scale_01 d_daughter, cluster(hid)

H2: reg revleftist d_daughter, cluster(hid)

H3: reg equality d_daughter, cluster(hid)

H4: reg attitude_scale_01 i.d_daughter##i.d_sex, cluster(hid)

Interpretation of results

(1) We will judge the p-value of the coefficient of having a daughter. We see a p-value below 10% as being in line with our hypothesis when the coefficient is in the expected direction (positive value). We decided on a p-value of 0.1 as we already posed restrictions on the minimum effect size and have a quite restricted sample.

(2) We also judge the effect size to see whether there is a substantial difference. For hypotheses H1, H3, and H4 (outcome variable ranging from 0 to 1), we argue that a coefficient below 0.05 is not substantial and the same for H2 (scale from 0 to 10) a coefficient below 0.5 is not substantial enough.

References

Kühhirt, M. (2012). Childbirth and the long-term division of labour within couples: How do substitution, bargaining power, and norms affect parents' time allocation in West Germany?. European sociological review, 28(5), 565-582.

Mills, M., & Begall, K. (2010). Preferences for the sex-composition of children in Europe: A multilevel examination of its effect on progression to a third child. Population studies, 64(1), 77-95.